LAB # 11 : Testing of Hypothesis

Introduction

This lab is concerned with the testing of the hypothesis. There are many problems in which, we must decide whether a statement concerning a parameter is true or false. Statistically speaking, we test a hypothesis about a parameter. We formulate a null hypothesis and an appropriate alternative hypothesis which we accept when the null hypothesis must be rejected. We need to specify the probability of a Type I error. If possible, desired, or necessary, also specify the probabilities of Type II errors for particular alternatives. The probability of a Type I error is also called the level of significance, and it is usually set at $\alpha = 0.05$ or $\alpha = 0.01$.
Based on the sampling distribution of an appropriate statistic, we construct a criterion for testing the null hypothesis against the given alternative. We calculate from the data the value of the statistic on which the decision is to be based. We decide whether to reject the null hypothesis or whether to fail to reject it.

Z- Test

The Z-test is used when we know the population's standard deviation σ and are comparing a sample mean to the population mean or comparing the means of two independent samples. For example, if the average height of a population is known, the Z-test can be used to determine whether a sample of individuals has a significantly different average height.
The Z-test is most reliable when working with large sample sizes, typically when $n \geq 30$. As the sample size increases, the sampling distribution of the sample mean approximates a normal distribution (Central Limit Theorem), making the Z-test a suitable choice for larger samples. For single sample with size n, the Z statistics is defined as

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}},$$

where $\overline{X}$is the sample mean, μ is mean of the population, and σ is population standard deviation.
For two samples, the value is given as

$$Z = \frac{(\overline{X_1} - \overline{X_2}) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}.$$

We provide one and two samples Z-test.

```
# one sample
install.packages("BSDA")
library(BSDA)
sample_data <- c(37,48,45,36,50,32,52,45,48,39)
result <-  z.test(sample_data, mu = 50,sigma.x=10,conf.level=0.98)
print(result)
```

```
# output
    One-sample z-Test
data:  sample_data
z = -2.1503, p-value = 0.03153
alternative hypothesis: true mean is not equal to 50
98 percent confidence interval:
   35.84344   50.55656
sample estimates:
mean of x
   43.2


# two sample
data_1 <- c(27, 24, 18, 29, 30,27,35,32,22,26)
data_2 <- c(23, 28, 20, 19, 35,23,19,32,21,23)
result <- z.test(data1,data2,mu=0,sigma.x=10,sigma.y=15)
print(result)
# output
    Two-sample z-Test
data:  data_1 and data_2
z = 0.47361, p-value = 0.6358
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
    -8.473514   13.873514
sample estimates:
mean of x mean of y
    27.0      24.3
```

t-Test

The t-Test is used to determine whether there is a significant difference between the means of two groups or between a sample and a known value.
For single sample with size n, the t statistics is defined as

$$t = \frac{\overline{X} - \mu}{s/\sqrt{n}},$$

where s is the sample mean.
One Sample t-test is given below.

```
sample_data <- c(52, 49, 53, 51, 48, 47, 54, 50)
result <- t.test(sample_data, mu = 50)
# output
   One Sample t-test
data:  sample_data
t = 0.57735, df = 7, p-value = 0.5818
alternative hypothesis: true mean is not equal to 50
95 percent confidence interval:
        48.45218    52.54782
```

```
sample estimates:
mean of x
        50.5
```

Two Sample t-test is given as follows.

```
set.seed(0)
data_1 <- rnorm(45, mean = 130, sd = 5)
data_2 <- rnorm(50, mean = 150, sd = 4)
t.test(data_1, data_2, var.equal = TRUE)
# output
   Two Sample t-test
data:  data_1 and data_2
t = -23.747, df = 93, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
       -21.22299  -17.94749
sample estimates:
mean of x mean of y
       130.2294   149.8146
```

F-Test

Fisher's F-test is used to compare the variances of two independent samples to determine if they come from populations with the same variance. It tests the ratio of two sample variances to see if they differ significantly.

```
sample_data1 <- c(10.2, 11.5, 9.8, 10.1, 12.3)
sample_data2 <- c(8.4, 9.2, 10.6, 7.9, 8.8)
result <- var.test(sample_data1, sample_data2)
# output
   F test to compare two variances
data:  sample_data1 and sample_data2
F = 1.0903, num df = 4, denom df = 4, p-value = 0.9352
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
        0.1135198     10.4718591
 sample estimates:
 ratio of variances
            1.090304
```

Exercises:

1. The lifespan (in hours) of two different types of parts used in a machine was recorded for eight samples of each type. The recorded lifespans are:

I :	130	120	135	128	123	136	130	119
II :	128	122	133	130	131	126	120	116

Test whether there is a significant difference in the mean lifespans of the two types of parts. Also, compute the 95% confidence interval for the difference in mean lifespans.

2. In an experiment, two different types of industrial furnaces are used to process metal parts, and the goal is to compare the variability in emissions produced by these furnaces. The data presents the emissions (in micrograms per cubic meter) measured for 10 different runs from each furnace:

Furnace I :	2.3	2.4	2.6	2.1	2.5	2.7	2.1	2.2	2.5	2.6
Furnace II :	1.5	1.8	2.0	1.9	1.7	1.6	1.8	2.1	2.0	1.6

Using a significance level of $\alpha = 0.05$, test whether there is a significant difference in the variances of emissions between the two furnaces. Also, compute the 95% confidence interval for the ratio of the variances.